Olympiad Test Paper — Social Science (Class 7) — Paper 4

Chapter 2: Understanding the Weather

Paper 4 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	2 — Understanding the Weather
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: confusing weather with climate, assuming pressure rises with altitude (it falls), swapping the jobs of thermometer/barometer/rain gauge/wind vane/anemometer/hygrometer, treating high humidity as a guarantee of rain, and mixing up units (mb / degrees / % / mm). Do not write on this paper — mark answers on your answer sheet.

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1. A travel guide reads: "Cherrapunji receives heavy rain almost every monsoon year, yet last 14 June was bright and dry there." Which pairing of the two facts to weather and climate is correct?

(A) "Heavy rain almost every monsoon" is the climate (a many-year pattern) and "bright and dry on 14 June" is the weather of that single day
(B) "Heavy rain almost every monsoon" is the weather and "bright and dry on 14 June" is the climate
(C) Both statements describe climate because both mention rain
(D) Both statements describe weather because both name a real place

2. A student argues: "Because Jaisalmer had three cool, rainy days in a row, its desert climate has now become wet." The single best correction is that:

(A) Three days are exactly enough to change a place's climate
(B) Rain can never fall in a place that has a desert climate
(C) A few days of unusual weather cannot redefine a climate, which is the average of weather over many years
(D) Weather and climate are identical, so the student is automatically right

3. Four claims are made about a place. Which one is the only valid statement about its CLIMATE rather than its weather?

(A) "Right now the sky is overcast and a light drizzle is falling"

(B) "The thermometer at the station reads 27 degrees Celsius this afternoon"

(C) "Over the last thirty years this town has been hot and dry in May and cool in December"

(D) "A strong gusty wind has been blowing since lunchtime today"

4. Why can two cities at almost the same latitude — one inland desert, one coastal — have very different CLIMATES even though they get sunlight for similar lengths of time?

(A) They cannot differ, because equal sunlight forces an identical climate

(B) The desert city is much closer to the Sun than the coastal city

(C) Climate is shaped by more than day-length; factors such as nearness to the sea give each place its own long-term pattern

(D) Climate is decided only by which city has more weather instruments

5. A meteorologist says, "To know whether a plane can take off in ten minutes I need one kind of information; to plan which crops suit this region over the coming years I need another." The correct pairing is:

(A) Take-off in ten minutes needs the climate; crop planning needs the weather

(B) Take-off in ten minutes needs the weather; long-range crop planning needs the climate

(C) Both decisions depend only on the weather of the next ten minutes

(D) Both decisions depend only on the thirty-year climate average

6. Nearly all clouds, rain and storms occur in the troposphere. Which statement BEST explains why this lowest layer is where weather happens, using only chapter ideas?

(A) It is the highest, thinnest layer, farthest from the Earth's surface

(B) It lies in empty space beyond the reach of the Sun's heat

(C) It is the layer where the oceans store most of their water

(D) It is the layer touching the ground, where the Sun-warmed surface, water vapour and rising air all interact

7. The troposphere reaches higher over the warm tropics than over the cold poles. Which cause-and-effect chain matches the chapter?

(A) Cold polar air expands more than warm tropical air, lifting the layer at the poles

(B) The tropics are physically much nearer the Sun, which pulls the layer outward

(C) Heavier rainfall at the poles thickens the layer there

(D) Warm air expands and rises, so it pushes the layer higher over the tropics, while cold air contracts and keeps the layer lower over the poles

8. A child says, "The atmosphere is just empty sky, so it cannot have weight or do anything." Why is this wrong, according to the chapter?

(A) The atmosphere is solid rock, which is why it has weight

(B) The atmosphere is made only of water, which falls as rain

(C) The atmosphere is a blanket of gases with real weight, and that weight pressing down is what we call atmospheric pressure

(D) The atmosphere has no weight, but it still causes wind by magic

9. Which single sequence correctly orders these layers from the one CLOSEST to the ground upward, given that weather happens in the lowest?

(A) Troposphere first (touching the ground), then the layers above it

(B) The layers above first, with the troposphere at the very top

(C) The oceans first, then the troposphere, then space

(D) Space first, then the troposphere, then the ground

10. A weather club is sorting cards. Which set contains ONLY genuine elements of weather and NO land or water feature?

(A) Temperature, rivers, wind, humidity (B) Pressure, mountains, rainfall, sunshine

(C) Wind, soil type, temperature, precipitation

(D) Temperature, atmospheric pressure, humidity, precipitation

11. Each weather element has its own unit. Which list pairs every element with its CORRECT unit?

(A) Temperature in millibars, pressure in degrees, rainfall in percent, humidity in millimetres

(B) Temperature in percent, pressure in millimetres, rainfall in degrees, humidity in millibars

(C) Temperature in millimetres, pressure in percent, rainfall in millibars, humidity in degrees

(D) Temperature in degrees, pressure in millibars, rainfall in millimetres, humidity in percent

12. A reading sheet lists three values stripped of their units: 1011, 32, 70. They belong to pressure, temperature and humidity. The only sensible assignment is:

(A) 1011 is the temperature in degrees, 32 is the pressure in millibars, 70 is the humidity in percent

(B) 1011 is the pressure in millibars, 32 is the temperature in degrees, 70 is the humidity in percent

(C) 1011 is the humidity in percent, 32 is the temperature in degrees, 70 is the pressure in millibars

(D) 1011 is the pressure in millibars, 32 is the humidity in percent, 70 is the temperature in degrees

13. A student insists that "the time of sunrise" should be added to the list of weather elements. The best reason to reject this is that:

(A) Sunrise time changes too fast to be a weather element

(B) Sunrise time is fixed by the Earth and Sun, not by the changing state of the air, so it is not a weather element

(C) Sunrise time is actually a form of precipitation

(D) Sunrise time is measured by the barometer, so it is pressure instead

14. Why do weather reports give each element a precise number on a shared scale instead of words like "warm" or "breezy"?

(A) Numbers make the weather itself change more slowly

(B) A measured value on a fixed scale means the same to everyone everywhere, so readings can be compared and combined

(C) Words are illegal to use in any weather report

(D) Numbers can only be read by the instruments, never by people

15. In a liquid-in-glass thermometer, what is the correct cause-and-effect chain when the day warms?

(A) Heat widens the glass tube, letting the liquid spread upward

(B) Warm air is pushed down into the tube, lifting the liquid

(C) Heat makes the liquid lighter, so it floats out of the bulb

(D) Heat makes the liquid expand, and with room to grow only up the narrow tube, the level rises

16. A log shows 25 degrees Celsius on Monday and 77 degrees Fahrenheit on Tuesday, and a student claims the thermometer is broken because the numbers differ so much. The correct reading is:

(A) 77 Fahrenheit must be far hotter than 25 Celsius because 77 is the bigger number

(B) Celsius and Fahrenheit are two scales for the same warmth, so different numbers are expected and the thermometer is fine

(C) Fahrenheit measures pressure, so the two readings cannot be compared

(D) The thermometer is faulty because one temperature must give one number

17. A station logs a maximum of 36 degrees and a minimum of 12 degrees in one day. Which single statement about the range AND mean is correct?

(A) The range is 24 degrees and the mean is 48 degrees

(B) The range is 48 degrees and the mean is 24 degrees

(C) The range is 24 degrees and the mean is 24 degrees

(D) The range is 12 degrees and the mean is 18 degrees

18. Five towns reported one day. Which town had the LARGEST daily temperature range? Town W: max 31, min 25. Town X: max 44, min 19. Town Y: max 27, min 14. Town Z: max 38, min 33.

(A) Town W, because its minimum is fairly high

(B) Town X, with a range of 25 degrees

(C) Town Z, because it has the second-highest maximum

(D) Town Y, because it has the lowest maximum

19. Reading the same five towns, which town stayed MOST STEADY (smallest range) across the day? Town W: max 31, min 25. Town X: max 44, min 19. Town Y: max 27, min 14. Town Z: max 38, min 33.

(A) Town X, because its numbers are the largest

(B) Town Y, because it has the lowest minimum

(C) Town W, with a range of 12 degrees (D) Town Z, with a range of only 5 degrees

20. Town A's daily range is 3 degrees and town B's is 19 degrees on the same date. What is the MOST that can safely be concluded?

(A) Town A was hotter than town B throughout the day

(B) Town B definitely received more rainfall than town A

(C) Town B's temperature swung far more between its daytime high and night-time low than town A's did

(D) Town A's average temperature must be lower than town B's

21. Two cities share the same MEAN daily temperature of 26 degrees, but city P has a range of 4 degrees and city Q a range of 20 degrees. What does this most likely tell us?

(A) City Q had a much greater difference between its day and night temperatures than city P, even though their averages match

(B) City Q must have been hotter than city P all day long

(C) City P must have received less rainfall than city Q

(D) The two cities cannot share a mean if their ranges differ

22. Precipitation is best defined as which of these, taking the chapter's full meaning?

(A) Any form of water — rain, snow, sleet or hail — that falls from the sky to the ground

(B) Only the rain that falls during the monsoon season

(C) The invisible water vapour held in the air

(D) The weight of all the air pressing down on the ground

23. Hail and dew are both linked to water in the air, but they differ sharply. Which contrast is correct?

(A) Both hail and dew fall from clouds as solid ice

(B) Hail is invisible vapour while dew is a strong wind

(C) Hail falls from the sky as hard balls of ice, while dew forms as droplets that settle on cool surfaces without falling from clouds

(D) Hail is measured in percent while dew is measured in millibars

24. A rain gauge reports "25 mm" for the day. Two gauges set side by side in the same yard, however, read 25 mm and 3 mm. What is the soundest conclusion?

(A) The 25 mm reading means it rained for exactly 25 minutes, and both gauges are fine

(B) The 25 mm reading means rain would lie 25 mm deep on flat ground, and the 3 mm gauge is probably blocked or faulty since both should read nearly the same

(C) The 25 mm reading is a volume of 25 millilitres, and the storm simply skipped one gauge

(D) Rainfall naturally falls to a tenth within one metre, so both readings are correct

25. Snow falls into an open rain gauge overnight and is still frozen at dawn. What is the correct step before reading the scale, and why?

(A) Let the snow melt to water first, then read the depth, because a rain gauge measures the depth of liquid water

(B) Read the depth of the solid snow directly, because snow and water take the same depth

(C) Tip the snow out and record zero, because frozen water does not count as precipitation

(D) Double the snow depth, because melting always halves the volume

26. Atmospheric pressure at a point is best understood as which of the following?

(A) The amount of invisible water vapour held in the air at that point

(B) The downward push of the weight of the whole column of air stacked above that point

(C) The speed at which the wind is crossing the ground at that point

(D) The temperature of the air high above that point

27. A mountaineer climbs from a coast at 1012 mb to a high pass. Which single statement correctly links her altitude, the pressure and her breathing?

(A) As she climbs, less air remains above her, so the pressure falls and the thinner air gives less oxygen per breath

(B) As she climbs, she nears the top of the sky, so the pressure rises and breathing gets easier

(C) The pressure stays exactly 1012 mb all the way up, so her breathing does not change

(D) The pressure jumps to zero at the pass, so there is no air at all to breathe

28. A hill station reads 680 mb while a coastal town reads 1012 mb at the same hour. A student offers four explanations. Which is correct?

(A) The hill station's barometer must be broken to read so low

(B) The coast is much colder that day, which raises its pressure

(C) The hill station sits much higher up, so less air lies above it and its pressure is lower

(D) The hill station had heavier rain, which lowered its pressure

29. Normal sea-level pressure is about 1013 mb. A barometer at a coastal station drops steadily to 985 mb over a few hours. The best interpretation is:

(A) Clear, calm and settled weather is now guaranteed

(B) The station has suddenly risen higher above sea level

(C) A depression is forming, and the falling pressure is an early warning that a storm may develop

(D) The air has become drier and safer, with no risk at all

30. Assertion: Atmospheric pressure is greater at the seashore than on a high mountain top. Reason: The seashore has more air stacked above it than the mountain top does. Choose the best option.

(A) The assertion is true but the reason is false

(B) Both the assertion and the reason are true, and the reason correctly explains the assertion

(C) The assertion is false but the reason is true

(D) Both are true, but the reason does not explain the assertion

31. Two travellers at different altitudes radio their barometer readings to compare. Traveller 1 at a valley reads 1005 mb; traveller 2 on a ridge reads 940 mb. Without any other information, what can be concluded?

(A) Traveller 2 is at a lower altitude, because lower pressure means lower ground

(B) Both travellers must be at exactly the same height, since air is everywhere

(C) Traveller 1 is higher up, because the larger pressure number means greater height

(D) Traveller 2 is at a higher altitude than traveller 1, because pressure falls as height increases

32. Wind is air in motion. Which single statement correctly states both WHY it blows and its DIRECTION between two regions?

(A) Air moves from high pressure to low pressure, so wind blows from the higher-pressure region toward the lower-pressure region

(B) Air moves from low pressure to high pressure, so wind blows from the lower toward the higher

(C) Air moves from cold to warm regardless of pressure, so wind ignores pressure entirely

(D) Air moves only straight upward, so there is no sideways wind between regions

33. Region G is at 1022 mb and region H beside it is at 998 mb. A second pair, region J at 1012 mb and region K at 1008 mb, lies the same distance apart. Comparing the two pairs, the wind will be:

(A) Stronger between J and K, because their smaller difference drives a faster wind

(B) Equal in both pairs, because wind speed does not depend on pressure difference

(C) Blowing from H to G and from K to J, that is from low pressure to high

(D) Stronger between G and H, because their larger pressure difference drives a faster wind

34. A weather station must record BOTH the direction the wind comes from AND how fast it blows. Which pairing of instruments to jobs is entirely correct?

(A) An anemometer for the direction, and a wind vane for the speed

(B) A barometer for the direction, and a thermometer for the speed

(C) A rain gauge for the direction, and a hygrometer for the speed

(D) A wind vane for the direction the wind comes from, and an anemometer for the wind's speed

35. At an airport a fabric wind sock hangs limp at dawn but by noon stretches out straight and level. What does the pilot correctly read from this change?

(A) The wind weakened from strong to calm, since a straight sock means little wind

(B) The air pressure over the runway rose during the morning

(C) The wind strengthened from near-calm to strong by noon

(D) Heavy rainfall began over the runway by noon

36. An anemometer's cups are barely turning while, at the same moment, a wind vane's arrow swings firmly to point south. The correct combined reading is:

(A) The wind is very strong, and the vane shows its speed as high

(B) The wind is gentle, and the slow cups show the wind's direction

(C) The air pressure is high, which is why the cups turn slowly

(D) The wind is gentle (low speed) and is coming from the south

37. Assertion: A wind vane and an anemometer can be swapped because they do the same job. Reason: A wind vane shows wind direction while an anemometer measures wind speed. Choose the best option.

(A) Both the assertion and the reason are true, and the reason explains it

(B) The assertion is false but the reason is true

(C) Both are true, but the reason does not explain the assertion

(D) The assertion is true but the reason is false

38. Humidity refers to which of the following, and is reported in which unit?

(A) The weight of the air above us, given in millibars

(B) The amount of water vapour in the air, given as a percentage

(C) The depth of rain collected, given in millimetres

(D) The speed of the wind, given in degrees

39. A hygrometer reads 95 % at noon and 30 % by mid-afternoon. Which statement about the air at those two times is correct?

(A) At noon the air was completely dry; by afternoon it was nearly saturated

(B) At noon it was definitely raining; by afternoon the temperature had reached 30 degrees

(C) Both readings mean the air held exactly the same amount of vapour

(D) At noon the air was very damp, nearly as full of vapour as it could hold; by afternoon it was dry and far from full

40. Two same-temperature days are compared. On Monday the relative humidity is 30 %, on Tuesday it is 90 %. On which day will wet washing dry FASTER, and why?

(A) Monday, because the drier air is far from full and can readily absorb more moisture from the clothes

(B) Tuesday, because the moist air pulls water out of the clothes faster

(C) Monday, because high pressure on that day squeezes water from the cloth

(D) Both days equally, because humidity has no effect on drying

41. On a muggy day a hygrometer reads 92 %, yet no rain falls all afternoon. Which explanation fits the chapter best?

(A) High humidity always brings rain, so the hygrometer must be faulty

(B) High humidity alone is not enough; the air must also cool enough for the vapour to condense before rain can form

(C) Humidity and rainfall have nothing to do with each other

(D) A reading of 92 % actually means the air is extremely dry

42. A weather station owns a thermometer, a barometer, a wind vane and an anemometer, but no hygrometer and no rain gauge. Which TWO weather elements can it NOT record?

(A) Humidity and rainfall (B) Temperature and pressure

(C) Wind direction and wind speed (D) Pressure and wind speed

43. Putting the instruments together, which single element-to-instrument pairing is INCORRECT?

(A) Temperature is measured by the thermometer

(B) Rainfall is measured by the rain gauge(C) Humidity is measured by the barometer

(D) Wind speed is measured by the anemometer

44. Before instruments existed, people watched ants carry eggs uphill or birds fly low to sense coming rain. What is the chief WEAKNESS of relying only on such nature's clues?

(A) They give precise numbers for pressure and humidity

(B) They are always perfectly accurate and never wrong

(C) They gather data from the whole country at once

(D) They are tied to one spot and give only vague, short-range hints rather than exact measurements

45. How does meteorology improve on simply reading nature's clues?

(A) It uses instruments to measure each element precisely and pools records from many places over time, so it can forecast ahead

(B) It relies on watching a single animal as its only guide

(C) It refuses to use any instruments and only studies the sky by eye

(D) It can describe only today and can never predict the future

46. An Automated Weather Station (AWS) is set up on a remote Himalayan ridge. Why is an AWS especially suited to such a place compared with an ordinary station?

(A) It needs an operator to read every instrument each hour by hand

(B) Its sensors record and store data by themselves, so it keeps working where it is hard or dangerous for a person to stay

(C) It works only inside large cities during the daytime

(D) It measures the region's thirty-year climate instead of its weather

47. A child claims, "Meteorologists just look at the sky and guess what the weather will be." Which correction is accurate?

(A) They measure each element with instruments, collect data from many places over time, and apply scientific methods

(B) They do indeed only guess from the look of the clouds

(C) They rely entirely on the behaviour of frogs and ants

(D) They simply copy whatever the newspaper printed yesterday

48. Day and night and the whole pattern of weather are ultimately driven by which single cause?

(A) The Moon's gravity pulling on the air

(B) The Sun's heat, which warms the land, sea and air unevenly

(C) The spinning of wind vanes at weather stations

(D) The barometers raising and lowering the pressure

49. Under one midday Sun, a bare sandy patch warms much more than a nearby pond, the air above the sand rises, and a breeze flows from the pond toward the sand. Which chain explains how the Sun causes this wind?

(A) The Sun heats the two surfaces unevenly, creating a pressure difference, and air flows from the higher to the lower pressure

(B) The Sun heats every surface exactly equally, so the air is pushed at random

(C) The Sun's rays physically blow the air sideways across the ground

(D) The Sun cools the sand, freezing the air so it cannot move

50. Why are the tropics generally warmer than the polar regions, given that all parts of Earth are at about the same distance from the Sun?

(A) The tropics are physically much closer to the Sun than the poles

(B) The poles receive far more hours of sunshine every day all year

(C) Sunlight strikes the tropics nearly overhead so its heat is concentrated, while it hits the poles at a slant and spreads thinly

(D) A stream of cold air from space pours only onto the tropics

51. On a clear day a town's temperature climbs from dawn, peaks in the early afternoon, and falls to its lowest just before the next dawn. This daily rise and fall is direct evidence that:

(A) The barometer controls the temperature throughout the day

(B) Humidity is what makes the Sun rise and set each day

(C) Wind speed decides whether it is day or night

(D) The Sun is the main heat source: temperature rises while it warms the surface and falls once that heating stops

52. Why does the lowest temperature of a clear day usually occur just before sunrise rather than at midnight?

(A) The Sun secretly heats the ground at midnight and stops before dawn

(B) The barometer pushes the temperature up again after midnight

(C) Humidity always peaks at midnight and warms the air until dawn

(D) The ground keeps losing heat through the whole night, so it is coolest after the longest stretch without the Sun, just before dawn

53. A forecast warns fishermen of an approaching cyclone so they stay ashore and warns a city to clear its drains before heavy rain. This shows the chief value of forecasts is that they:

(A) Stop the cyclone or the flood from ever forming

(B) Make the weather change less often than it used to

(C) Remove the need for any weather instruments

(D) Give people and authorities time to prepare and stay safe before severe weather arrives

54. In India, which body gathers weather data nationwide and issues forecasts and cyclone warnings, and is therefore the correct one to credit in a school report?

(A) The National Rain Gauge Office (B) The Department of Climate History

(C) The India Meteorological Department (D) The Indian Wind Vane Society

55. To track a thermometer, rain gauge, barometer, wind vane, anemometer and hygrometer together in one place, reading them at fixed times, what should you set up?

(A) A weather station (B) A climate map

(C) A wind sock (D) A depression

56. A weather log for one afternoon reads: 1 p.m. pressure 1011 mb, humidity 50 %, light wind, clear sky; 2 p.m. pressure 1003 mb, humidity 68 %, freshening wind, cloud building; 3 p.m. pressure 994 mb, humidity 86 %, strong wind, dark cloud. The best forecast for the evening is:

(A) Clear, calm and dry weather will continue (B) A cold, cloudless evening with no wind

(C) Wet, stormy weather is approaching (D) No change at all from the 1 p.m. conditions

57. In that same log the pressure falls 1011 to 1003 to 994 mb. Why is this steady fall an early warning, and what does it most likely mark?

(A) Falling pressure marks the arrival of clear, settled fair weather

(B) Falling pressure often marks the approach of a depression, the seed of a storm

(C) Falling pressure means the station is rising higher above sea level

(D) Falling pressure marks the complete end of all wind in the area

58. A weather diary records, in order, these events one stormy afternoon: the barometer began to fall, then the wind picked up, then dark clouds gathered, then rain began. Which arrangement keeps this cause-and-effect order from earliest sign to final result?

(A) Rain first, then gathering cloud, then strengthening wind, then falling pressure

(B) Strengthening wind first, then rain, then falling pressure, then gathering cloud

(C) Falling pressure first, then strengthening wind, then gathering cloud, then rain

(D) Gathering cloud first, then falling pressure, then rain, then strengthening wind

59. Identify the ODD ONE OUT: three of the following instruments report a property of moving or pressing air, while one reports water that has fallen. Which is the odd one out?

(A) Rain gauge
(B) Barometer
(C) Wind vane
(D) Anemometer

60. Which single statement chains together the chapter's ideas with EVERY part correct?

(A) Pressure falls with altitude, is read by a barometer in millibars, and a sharp fall can warn of a storm

(B) Pressure rises with altitude, is read by a thermometer, and a fall means clear skies

(C) Humidity is read by an anemometer in degrees, and 100 % means very dry air

(D) Rainfall is read by a wind vane in percent, and high humidity always brings rain

Solutions — Social Science (Class 7), Chapter 2: Understanding the Weather — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	D	21	A	31	D	41	B	51	D
2	C	12	B	22	A	32	A	42	A	52	D
3	C	13	B	23	C	33	D	43	C	53	D
4	C	14	B	24	B	34	D	44	D	54	C
5	B	15	D	25	A	35	C	45	A	55	A
6	D	16	B	26	B	36	D	46	B	56	C
7	D	17	C	27	A	37	B	47	A	57	B
8	C	18	B	28	C	38	B	48	B	58	C
9	A	19	D	29	C	39	D	49	A	59	A
10	D	20	C	30	B	40	A	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Climate is the average pattern repeated over many years, so 'heavy rain almost every monsoon' is climate; the condition of one named day, 14 June, is weather. The first wrong option swaps the two labels. Naming rain does not make a statement climate, and naming a place does not make a statement weather; the deciding factor is whether the description is of one moment or of a long-term average.

2. (C) Climate is built from weather records over many years, so three unusual days cannot overturn it. The first wrong option treats a tiny sample as enough to fix climate; the second wrongly says deserts never get rain, when a dry climate can still have occasional rainy weather; the third falsely equates weather and climate, which differ in time-scale.

3. (C) Climate is the long-term average pattern, so the thirty-year hot-May / cool-December description is climate. The other three each report the atmosphere's state at one moment — present drizzle, this afternoon's reading, today's gusts — which is weather, not climate.

4. (C) Climate is the long-term average of a place's weather and is influenced by many things, including how near the sea a place lies, so two cities at one latitude can differ. Equal day-length does not force an identical climate; both cities are at essentially the same distance from the Sun; and the number of instruments only measures the weather, it does not create the climate.

5. (B) A take-off depends on conditions right now — the weather; choosing crops for years ahead depends on the long-term average — the climate. The first wrong option reverses the two; the others wrongly collapse both decisions onto a single time-scale, when the near-term flight needs weather and the multi-year planning needs climate.

6. (D) The troposphere is the lowest layer, touching the ground, so it holds the warmed surface air and the water vapour that form clouds and rain. The first wrong option describes a high, thin layer, not the lowest one; the second places it beyond the atmosphere in space; the third confuses an atmospheric layer with the oceans.

7. (D) Warm air expands and rises, lifting the troposphere over the tropics, whereas cold air shrinks and lowers it over the poles. The first wrong option reverses the behaviour of warm and cold air; the second wrongly claims the tropics are closer to the Sun (all latitudes are about the same distance); the third invents rainfall as the cause of layer thickness.

8. (C) The atmosphere is the blanket of gases around Earth; those gases have weight, and that weight pressing down is atmospheric pressure. The first wrong option calls air solid rock; the second says it is only water; the third denies that air has weight, when its weight is exactly what creates pressure and drives wind.

9. (A) The troposphere is the lowest layer, touching the ground, so it comes first when ordering upward; that is why weather forms there. The first wrong option puts the troposphere at the top instead of the bottom; the second inserts the oceans as an atmospheric layer; the third begins with space, which is the outermost region, not the closest to the ground.

10. (D) Elements of weather describe the changing state of the air — temperature, pressure, humidity and precipitation all qualify. Each wrong set smuggles in a land or water feature: rivers, mountains and soil type are features of the land or its water, not conditions of the atmosphere, so those sets are mixed and invalid.

11. (D) Temperature is read in degrees, pressure in millibars, rainfall depth in millimetres and humidity in percent. Each wrong option scrambles these units onto the wrong elements, so although every unit named is real, none of the wrong sets matches the elements correctly.

12. (B) Sea-level pressure runs near 1000 mb, a daytime temperature is a couple of dozen degrees, and humidity is a percentage that cannot exceed 100 — so 1011 mb / 32 degrees /

70 % is the only physically possible fit. Each wrong option produces an impossible figure, such as a humidity of 1011 % or a pressure of only 70 mb at the surface.

13. (B) Weather elements describe the changing condition of the atmosphere — temperature, pressure, humidity, wind, precipitation; sunrise time is set by astronomy, not by the air's state, so it does not belong. The first wrong option misjudges sunrise time as fast-changing weather; the second calls it precipitation; the third claims a barometer measures it, when a barometer measures pressure.

14. (B) A number on a shared scale carries the same meaning for everyone, allowing readings from many places to be compared and pooled into forecasts. The first wrong option imagines numbers alter the weather; the second invents a ban on words; the third wrongly says people cannot read the values, when the whole point is that anyone can read and compare them.

15. (D) Warming makes the liquid expand, and because the only space to expand into is up the thin bore, the column rises. The first wrong option blames the glass widening; the second imagines outside air entering the sealed tube; the third treats the rise as floating, when it is expansion taking up space.

16. (B) Temperature can be reported on either scale, and the same warmth simply shows a different number on each, so nothing is broken. The first wrong option assumes the larger number means hotter, ignoring the scale; the second wrongly says Fahrenheit is a pressure scale; the third denies that one temperature can legitimately have two scale readings.

17. (C) Range is max minus min, 36 - 12 = 24; mean is (max + min) / 2 = (36 + 12) / 2 = 24. Here the range and mean happen to be equal. The first wrong option uses the un-halved sum (48) as the mean; the second uses the un-halved sum as the range; the third miscomputes both figures.

18. (B) Ranges are W = 6, X = 25, Y = 13, Z = 5, so the largest gap, 25 degrees, belongs to X. A high minimum (W), a high maximum (Z) or a low maximum (Y) does not by itself set the range — only the gap between a town's own high and low does, and Z and W in fact have the smallest swings.

19. (D) Ranges are W = 6, X = 25, Y = 13, Z = 5, so Z varied least. Town X's large numbers actually give the biggest swing; Y's low minimum does not mean a small range; and W's range is 6, not 12, so it is not the steadiest.

20. (C) Range measures only the gap between a place's own high and low, so a larger range means a bigger day-to-night swing — nothing more. It says nothing about which town was hotter, which got more rain, or which had the higher average, since two places can share a mean yet differ widely in range.

21. (A) A larger range means a bigger swing between high and low, so Q's day and night differed far more than P's, yet both can average 26. The first wrong option confuses range with how hot it was; the second invents a rainfall link; the third wrongly claims equal means are impossible with different ranges, when range and mean are independent.

22. (A) Precipitation is water that falls from the sky in any form, whether liquid rain or frozen snow, sleet or hail. The first wrong option limits it to monsoon rain only; the second describes humidity (invisible vapour), which has not yet fallen; the third describes atmospheric pressure, an unrelated element.

23. (C) Hail is a form of precipitation — solid ice pellets that fall from the sky — while dew gathers as droplets on cool surfaces and does not fall as precipitation. The first wrong option wrongly makes dew fall as ice; the second confuses hail with vapour and dew with wind; the third assigns nonsensical units, since these are not measured in percent or millibars.

24. (B) A rain-gauge figure is a depth — 25 mm is how deep the water would stand on level ground — and two gauges in one spot should agree, so a large gap points to a blocked or faulty gauge. The first wrong option treats mm as minutes; the second treats it as a volume in millilitres; the third invents a rule that rain drops tenfold within a metre, which does not happen over a single yard.

25. (A) A rain gauge measures liquid-water depth, so frozen snow must be melted and then read. The first wrong option reads solid snow directly, but snow is far less dense than water so its depth overstates the figure; the second discards real precipitation as zero; the third applies an invented doubling rule with no basis.

26. (B) Air has weight, and pressure at a place is the push of the entire column of air above it bearing down. The first wrong option describes humidity (vapour content); the second describes wind speed; the third describes temperature — none of these is the weight of the overlying air.

27. (A) Pressure comes from the weight of air above, which decreases with height, so pressure falls and the thinner air holds less oxygen, leaving her breathless. The first wrong option reverses the trend; the second wrongly keeps pressure constant; the third claims a vacuum at the pass, when the air merely thins rather than vanishing.

28. (C) Pressure falls with altitude because less air remains overhead, so a lower hill-station reading is exactly expected. The first wrong option blames a broken instrument; the second invents a temperature cause; the third invents a rainfall cause — the height difference alone accounts for the lower reading.

29. (C) A sharp fall well below 1013 mb marks a depression, which can grow into a storm, so falling pressure is an early warning. The first wrong option links the fall to fair weather, which actually goes with rising pressure; the second imagines the fixed station changing altitude in hours; the third treats a deepening low as safe, when it signals unsettled, possibly stormy weather.

30. (B) Pressure is higher at the seashore because more air lies above a low place than above a high one, so both statements are true and the reason explains the assertion. The first wrong option falsely calls the reason wrong; the second falsely denies the true assertion; the third wrongly says the link is unexplained, when the weight of overlying air is exactly the cause.

31. (D) Pressure decreases with altitude, so the lower reading (940 mb) belongs to the higher traveller, number 2. The first wrong option reverses height and pressure; the second wrongly claims equal heights despite different readings; the third assumes the bigger pressure number means greater height, which is backwards.

32. (A) Wind is air flowing from high pressure toward low pressure. The first wrong option reverses the direction; the second denies the role of pressure, when pressure differences are what drive wind; the third says air only rises, ignoring the horizontal flow that pressure differences create.

33. (D) The greater the pressure difference over a given distance, the faster the wind, so the 24 mb gap (G-H) drives a stronger wind than the 4 mb gap (J-K). The first wrong option claims the smaller gap is faster; the second denies that pressure difference affects speed; the third reverses the direction, since wind blows high-to-low (G to H, J to K).

34. (D) A wind vane shows the direction the wind comes from, and an anemometer's spinning cups measure wind speed. The first wrong option swaps the two roles; the second uses a barometer and thermometer, which measure pressure and temperature; the third uses a rain gauge and hygrometer, which measure rainfall and humidity.

35. (C) A wind sock hangs limp in calm air and stretches out level only when the wind is strong, so a limp-to-level change over the morning means the wind strengthened from near-calm to strong. One wrong option reverses the meaning of a level sock (claiming it shows calm); another confuses the sock with a barometer reading pressure; the last confuses it with a rain gauge measuring rainfall.

36. (D) Slowly turning cups mean a low wind speed, while the vane's arrow points to where the wind comes from — here the south. The first wrong option misreads slow cups as strong wind and gives the vane a speed job; the second wrongly says the cups give direction; the third confuses the anemometer with a barometer.

37. (B) The two instruments measure different properties — direction versus speed — so they cannot be swapped, making the assertion false; the reason states their true, distinct jobs. The reason actually shows why the assertion is wrong, so any option treating the assertion as true is incorrect, and the reason itself is not false.

38. (B) Humidity is the water vapour present in the air, and relative humidity is reported as a percentage. The first wrong option describes pressure in millibars; the second describes rainfall in millimetres; the third describes wind speed — none of these is the air's vapour content.

39. (D) A high relative humidity (95 %) means the air is nearly saturated and damp, while a low one (30 %) means it is dry — so the noon air was damp and the afternoon air dry. The first wrong option reverses damp and dry; the second confuses a humidity figure with rain and with temperature; the third wrongly claims the differing readings mean equal vapour.

40. (A) Low humidity means the air still has room to take up vapour, so moisture leaves the clothes quickly — Monday dries faster. The first wrong option claims moist air dries clothes faster, which is backwards; the second invents pressure as the cause when humidity is the factor; the third denies humidity's clear effect on drying.

41. (B) Rain needs the vapour to condense, which usually requires cooling, so high humidity makes rain likely but does not guarantee it. The first wrong option assumes humidity must bring rain and blames the instrument; the second denies any link, when humidity supplies the moisture; the third misreads 92 % as dry, when it means very moist.

42. (A) Humidity needs a hygrometer and rainfall needs a rain gauge, both of which are missing, so those two cannot be recorded. The first wrong pair (temperature, pressure) is covered by the thermometer and barometer; the others (wind direction and speed) are covered by the wind vane and anemometer, so each of the wrong options names elements the station can in fact measure.

43. (C) Humidity is measured by a hygrometer, not a barometer (which measures pressure), so that pairing is the wrong one. Temperature-thermometer, rainfall-rain gauge and wind-speed-anemometer are all correctly matched, so the incorrect pairing is the humidity-barometer one.

44. (D) Nature's clues are local and qualitative, warning only roughly and only for the near future — that is their main limitation. The first wrong option credits them with precise numbers they cannot give; the second claims perfect accuracy; the third claims nationwide coverage, but a single spot's animals cannot represent a whole country.

45. (A) Meteorology is the systematic, scientific study of weather: it takes exact instrument readings and combines data from many stations over time, which is what lets it predict. The first wrong option reduces it to one animal; the second denies it uses instruments; the third denies it can forecast, when gathering data over time is exactly what enables prediction.

46. (B) An AWS senses and logs readings automatically, so it can run unattended in hard-to-reach or risky places. The first wrong option contradicts its automatic nature by requiring a human reader; the second wrongly limits it to cities and daylight; the third confuses it with a climate record, when it measures present weather like any station.

47. (A) Meteorology is systematic and instrument-based, gathering data across places and time and reasoning scientifically — not guessing. The first wrong option accepts the guessing claim; the second reduces it to watching animals; the third makes it mere copying, when it is built on measured data and method.

48. (B) The Sun's energy heats land, sea and air unevenly, setting temperature differences, pressure differences, winds and the water cycle in motion — so the Sun drives the weather. The Moon's pull mainly affects tides; wind vanes and barometers only measure the weather, they do not cause it.

49. (A) Different surfaces warm at different rates, so the air above them differs in temperature and pressure, and air flows from high to low pressure — that flow is the breeze. The first wrong option claims equal heating, which would create no difference; the second imagines rays pushing air directly; the third describes cooling and freezing, the opposite of the midday warming described.

50. (C) Direct, near-overhead sunlight concentrates its energy on a small area in the tropics, while the same beam strikes the poles at a low angle and spreads out, so the tropics warm more. The first wrong option wrongly claims the tropics are closer to the Sun; the second wrongly gives the poles more daily sun; the third invents cold space-air singling out the tropics.

51. (D) Temperature climbs while the Sun heats the surface and falls after that heating ends at night, reaching its lowest near dawn — clear evidence the Sun drives the day's heat. The first wrong option gives the barometer a control role it lacks; the second and third confuse humidity and wind, which describe other elements, with the cause of day and night.

52. (D) With the Sun absent all night the surface keeps losing heat, so it is coldest after the longest gap without sunlight, which is just before dawn. The first wrong option invents midnight heating by the Sun; the second gives the barometer a heating role; the third claims humidity warms the air, when it is the lack of solar heating that lets the temperature keep falling.

53. (D) Forecasts cannot change the weather, but the advance notice lets people protect lives and property. The first wrong option claims they prevent the storm itself; the second claims they slow how often weather changes; the third says they replace instruments, when forecasts actually depend on instrument data.

54. (C) The India Meteorological Department is the official agency that collects weather data and issues India's forecasts and warnings. The other three names are invented and have no such national role, so only the India Meteorological Department is correct.

55. (A) A weather station gathers the full set of instruments in one site and records them at set times so all elements are tracked together. A climate map shows long-term averages, a wind sock only indicates wind, and a depression is a region of low pressure — none of these brings the instruments together for regular readings.

56. (C) Falling pressure, rising humidity, strengthening wind and thickening dark cloud together point to an approaching storm with rain. The other options contradict the trends in the log: clear and calm, cloudless, or unchanged conditions all clash with the dark cloud building and the steadily dropping pressure.

57. (B) A steady pressure drop signals air moving toward a low — a depression — which can deepen into a storm, so it is an early warning. The first wrong option links the fall to fair weather, which goes with rising pressure; the second imagines a fixed station changing altitude within an afternoon; the third predicts no wind, when falling pressure usually brings stronger wind.

58. (C) A depression shows itself first as falling pressure, which drives stronger wind, which is followed by building cloud and finally rain — so the falling-pressure-first order is correct. Each wrong option breaks the chain by putting a later result (rain or cloud) or the driven wind before the early-warning pressure fall.

59. (A) A rain gauge measures the depth of fallen water (precipitation), making it the odd one out. The barometer measures the pressing weight of air, while the wind vane and anemometer measure the direction and speed of moving air — all three concern the air itself rather than fallen water.

60. (A) Pressure does decrease with height, is measured by a barometer in millibars, and a steep fall signals a possible storm — every part is right. The first wrong option reverses the altitude trend and mis-assigns the thermometer; the second mis-assigns the anemometer and degrees to humidity and reverses the meaning of 100 %; the third mis-assigns the wind vane and percent to rainfall and wrongly guarantees rain from humidity.